

Dumbbell with viscosity — Solution

X26 (2026) — English translation

A1

0.60 pts

Find the distance x_1 from the end of the dumbbell m_1 to the attachment point.

With a fixed orientation of the rotation axis relative to the dumbbell, the Huygens-Steiner theorem can be used to minimize the moment of inertia: for the rotation axis passing through an arbitrary point P, the moment of inertia J_P is related to the moment of inertia J_O , where O is the center of mass of the system, as

$$J_P = J_O + M \cdot OP^2.$$

Thus, in the case of minimum J_P correct P = O. For calculation distance from edge m_2 we use the definition of the center of mass:

Answer: Answer:

$$x_2 = \frac{m_1}{m_1 + m_2} l.$$

A2

0.20 pts

Find the moment of inertia J .

Let's find the distance x_1 from the heavy end of the dumbbell to its center of mass:

$$x_1 = \frac{m_2}{m_1 + m_2} l.$$

Let's use the definition of the moment of inertia:

Next, for convenience, we introduce the reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Then

$$J_O = \mu l^2.$$

Answer: Answer:

$$J_O = m_1 x_1^2 + m_2 x_2^2 = \frac{m_1 m_2}{m_1 + m_2} l^2.$$

A3

0.30 pts

Calculate the angular acceleration ε of the dumbbell at time $t = 0$.

In the case of a fixed axis of rotation, we can write the equation of rotational motion in the form

$$J \vec{\varepsilon} = \sum_i \vec{M}_i.$$

In the case of flat motion direction moment force and angular acceleration очевидны, thus that further rewrite equation in скалярном form. Согласно condition, external вынуждающий moment constant and equal $M_{\text{outer}} = F \cdot x_1$, and for zero angular velocity friction force be absent. Substitute expression for x_1 , we have

Answer: Answer:

$$\varepsilon = \frac{m_2}{m_1 + m_2} \frac{F l}{\mu l^2} = \frac{F}{m_1 l}.$$

A4

0.60 pts

Calculate the steady-state angular velocity ω_{lim} of the dumbbell rotation after a long time.

The frictional moment M_{fric} arises due to the linear velocities v_1, v_2 of the two point ends of the rod:

$$v_1 = \omega x_1; \quad v_2 = \omega x_2 \Rightarrow f_1 = -\gamma \omega x_1; \quad f_2 = -\gamma \omega x_2.$$

Очевидно, that moment direct против вынуждающего constant moment. Write it:

$$M_{\text{fric}} = f_1 x_1 + f_2 x_2 = -\gamma(x_1^2 + x_2^2)\omega.$$

Now we can write down the equation of rotational motion at an arbitrary moment of time:

$$\mu l^2 \varepsilon = \mu l^2 \frac{d\omega}{dt} = F x_1 - \gamma(x_1^2 + x_2^2)\omega \Leftrightarrow \mu l^2 \frac{d\omega}{dt} + \gamma(x_1^2 + x_2^2)\omega = F x_1.$$

Solution this equation даётсся following formula:

$$\omega(t) = \frac{F x_1}{\gamma(x_1^2 + x_2^2)} \left(1 - e^{-\frac{\gamma(x_1^2 + x_2^2)}{\mu l^2} t} \right) = \frac{F m_2(m_1 + m_2)}{\gamma(m_1^2 + m_2^2)l} \left(1 - e^{-\frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2)m_1 m_2} t} \right).$$

Further find ω_{lim} as предел $\omega(t)$ for $t \rightarrow \infty$, or from differential equation with condition $\dot{\omega} = 0$.

Answer: Answer:

$$\omega_{\text{lim}} = \lim_{t \rightarrow \infty} \omega(t) = \frac{m_2(m_1 + m_2)}{m_1^2 + m_2^2} \frac{F}{\gamma l}.$$

A5

0.50 pts

Find the moment of time τ at which the angular velocity is $\omega_{\text{lim}}/2$.

According to the explicit formula $\omega(t)$, obtained by solving the previous paragraph, we have the answer for the characteristic time τ (the time during which the exponent will significantly change its value).

Answer: Answer:

$$\tau = \frac{(m_1 + m_2) \cdot m_1 m_2}{\gamma(m_1^2 + m_2^2)} \ln(2).$$

B1

0.20 pts

Calculate the angular acceleration ε of the dumbbell at time $t = 0$.

The equation of rotational motion of a system with mass m for an axis moving translationally looks like

$$J \vec{\varepsilon} = \vec{M}_{\text{outer}} + m [\vec{v}_c, \vec{v}_p],$$

where $\vec{v}_p$ — (поступательная) velocity axis, and $\vec{v}_c$ — velocity of the center of mass. Thus, for move axis, coincide with center mass, equation of motion thus same take form

$$J \vec{\varepsilon} = \vec{M}_{\text{outer}},$$

and the answer for the angular acceleration of the previous part of the problem remains valid.

Answer: Answer:

$$\varepsilon = \frac{F}{m_1 l}.$$

B2

0.20 pts

How long will it take T_n for the n th revolution of the dumbbell?

In this part of the problem there is no friction, so the external moment is always equal to the driving moment, so it is constant. The dumbbell rotates uniformly accelerated from a position with zero initial angular velocity. The dependence $\omega(t)$ is given by the equality

$$\omega(t) = \varepsilon t,$$

and rotation angle from initial position, естественно,

$$\varphi = \frac{\varepsilon t^2}{2}.$$

The dumbbell will make n full revolutions in time t_n such that

$$2\pi n = \frac{\varepsilon t_n^2}{2} \Rightarrow t_n = 2\sqrt{\frac{\pi n}{\varepsilon}}.$$

It is clear then, that time n -ого оборота can be obtain equality

$$T_n = t_n - t_{n-1}.$$

Answer: Answer:

$$T_n = 2\sqrt{\frac{\pi m_1 l}{F}} (\sqrt{n} - \sqrt{n-1}) = 2\sqrt{\frac{\pi m_1 l}{F}} \frac{1}{\sqrt{n} + \sqrt{n-1}} \approx \sqrt{\frac{\pi}{n}} \frac{m_1 l}{F}.$$

B3

1.00 pts

After a long time, the speed of the dumbbell's center of mass becomes almost constant. Find the vector $\vec{u}$ of the steady-state velocity of the center of mass. In response, write down the x - and y -components of the speed $\vec{u}$.

It is clear that

$$u_x = \int_0^\infty \dot{v}_x dt = \frac{F}{m} \int_0^\infty -\sin\left(\frac{\varepsilon t^2}{2}\right) dt = -\frac{F}{m} \sqrt{\frac{2}{\varepsilon}} \int_0^\infty \sin(\xi^2) d\xi = -\frac{F}{m} \sqrt{\frac{2}{\varepsilon}} \sqrt{\frac{\pi}{8}}.$$

That is

$$u_x = -\frac{F}{2m} \sqrt{\frac{\pi}{\varepsilon}}.$$

Similarly,

$$u_y = \frac{F}{2m} \sqrt{\frac{\pi}{\varepsilon}}.$$

Answer: Answer:

$$\vec{u} = \frac{\sqrt{\pi m_1 F l}}{2(m_1 + m_2)} \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \quad (1)$$

C1

1.50 pts

Determine the final displacement vector $\vec{s}$ of the center of mass of the dumbbell after it stops. The answer must be presented in the form of components in the coordinate system shown in this figure.

Let's start counting the angle of deviation φ from the vertical in the direction of the minimum disturbance of the dumbbell. Let us write the theorem about the motion of the

center of mass in coordinate form:

$$\begin{cases} m\dot{v}_x = -2\gamma\dot{x} + 2\gamma\Delta \cos(\varphi)\omega; \\ m\dot{v}_y = -2\gamma\dot{y} - 2\gamma\Delta \sin(\varphi)\omega. \end{cases} \quad (2)$$

Now note that in this system we can get rid of the time derivative and relate changes in velocities, coordinates and rotation angle:

$$\begin{cases} \frac{m}{\gamma}dv_x = -2dx + 2\Delta \cos(\varphi)d\varphi; \\ \frac{m}{\gamma}dv_y = -2dy - 2\Delta \sin(\varphi)d\varphi. \end{cases} \Rightarrow \begin{cases} \frac{m}{\gamma}(v_x - v_{x_0}) = -2(x - x_0) + 2\Delta(\sin(\varphi) - \sin(\varphi_0)); \\ \frac{m}{\gamma}(v_y - v_{y_0}) = -2(y - y_0) + 2\Delta(\cos(\varphi) - \cos(\varphi_0)). \end{cases} \quad (3)$$

Let's take into account the initial conditions for our movement:

$$\begin{cases} v_{x_0} = 0; \\ v_{y_0} = 0; \\ x_0 = 0; \\ y_0 = 0; \\ \varphi_0 = 0; \end{cases} \quad (4)$$

and «final conditions»:

$$\begin{cases} \varphi = \pi; \\ v_x = 0; \\ v_y = 0. \end{cases} \quad (5)$$

Substituting them into the system, we have

$$\begin{cases} 0 = -2x + 0; \\ 0 = -2y + 2\Delta(-1 - 1). \end{cases} \quad (6)$$

From here

Answer: Answer:

$$\vec{s} = \begin{pmatrix} 0 \\ -2\Delta \end{pmatrix}$$

C2

0.30 pts

Determine how much heat was released during the entire movement of the dumbbell.

Let's use the law of conservation of energy and connect the work of potential forces (gravity and Archimedes' force) and dissipated energy Q :

$$(m_1 + m_2)g \cdot 2\Delta + F_a \cdot 0 = Q.$$

Answer: Answer:

$$Q = (m_1 - m_2)gl.$$

D1

0.60 pts

At an arbitrary moment in time, write down the total moment $\vec{M}_{out}$ of the external forces relative to the center of mass of the dumbbell. Express it in terms of the center of mass velocity $\vec{v}$, angular velocity $\vec{\omega}$, $\vec{\Delta}$, $\vec{g}$ and system parameters.

Let's write down the total moment of force $\vec{M}_{out}$ for the dumbbell relative to its center of mass:

$$\vec{M}_{out} = -\gamma \cdot [(\vec{r}_1 + \vec{r}_2) \times \vec{v}] + [\vec{\Delta} \times \vec{F}_a] - \gamma \cdot [\vec{r}_1 \times [\vec{\omega} \times \vec{r}_1]] - \gamma \cdot [\vec{r}_2 \times [\vec{\omega} \times \vec{r}_2]]$$

Obtain answer, transform vector product taking into account $\vec{r}_1 + \vec{r}_2 = 2\vec{\Delta}$:

$$\vec{M}_{\text{out}} = -2\gamma \cdot [\vec{\Delta} \times \vec{v}] + [\vec{\Delta} \times \vec{F}_a] - \gamma \cdot \vec{\omega} \cdot (r_1^2 + r_2^2).$$

Answer: Answer:

$$\vec{M}_{\text{out}} = -2\gamma \cdot [\vec{\Delta} \times \vec{v}] + [\vec{\Delta} \times \vec{F}_a] - \gamma \cdot \vec{\omega} \cdot (r_1^2 + r_2^2).$$

D2

0.20 pts

Show that under the conditions of small damping and $\varphi_0 \ll 1$ in the equation of rotational motion we can neglect the moment caused by the translational motion of the center of mass.

Let us write the theorem on the motion of the center of mass for a dumbbell:

$$(m_1 + m_2) \cdot \dot{\vec{v}} = -2\gamma \vec{v} - 2\gamma \cdot [\vec{\omega} \times \vec{\Delta}].$$

Integrate it by time:

$$(m_1 + m_2) \vec{v} = -2\gamma \vec{S} - 2\gamma \cdot \int_0^t [\vec{\omega} \times \vec{\Delta}] dt$$

First term in theorem of motion of the center of mass always directed против $\vec{v}$, therefore estimate value v сверху:

$$v < \frac{2\gamma}{m_1 + m_2} \cdot \Delta \varphi,$$

From which in force smallness damping, first term in write выше total moment force, turn out order γ^2 , and third — order $\gamma \Rightarrow$ moment caused by the translational movement of the center mass, can be neglected.

D3

0.50 pts

Write down the equation of rotational motion. Taking into account the smallness of the damping and the condition $\varphi_0 \ll 1$, you obtain the equation of damped oscillations of the angle formed by the dumbbell with the vertical:

$$\ddot{\varphi} + A\dot{\varphi} + B\varphi = 0.$$

Determine coefficient A and B .

From the results of points D1 and D2 we write the equation of rotational motion for the dumbbell relative to its center of mass:

$$I\vec{\varepsilon} = [\vec{\Delta} \times \vec{F}_a] - \gamma \cdot \vec{\omega} \cdot (r_1^2 + r_2^2)$$

Substitute in given equation геометрические parameter from part A, $\varepsilon = \ddot{\varphi}$, $\omega = \dot{\varphi}$ and спроецируем it on axis, perpendicular plane figure:

$$\ddot{\varphi} + \frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2) \cdot m_1 m_2} \dot{\varphi} + \frac{\Delta(m_1 + m_2)^2 g}{m_1 m_2 l^2} \varphi = 0$$

. Obtain итоговый answer:

$$A = \frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2) \cdot m_1 m_2}$$

$$B = \frac{\Delta(m_1 + m_2)^2 g}{m_1 m_2 l^2}.$$

Answer: Answer:

$$A = \frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2) \cdot m_1 m_2}$$

$$B = \frac{\Delta(m_1 + m_2)^2 g}{m_1 m_2 l^2}.$$

D4

0.20 pts

Determine the period T of the dumbbell oscillations.

In the previous paragraph we obtained the equation of damped oscillations, for which the period is calculated using the formula:

$$T = \frac{2\pi}{\sqrt{B - A^2/4}} = \frac{2\pi}{\sqrt{\frac{\Delta(m_1 + m_2)^2 g}{m_1 m_2 l^2} - \frac{1}{4} \cdot \left(\frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2) \cdot m_1 m_2} \right)^2}}.$$

Answer: Answer:

$$T = \frac{2\pi}{\sqrt{\frac{\Delta(m_1 + m_2)^2 g}{m_1 m_2 l^2} - \frac{1}{4} \cdot \left(\frac{\gamma(m_1^2 + m_2^2)}{(m_1 + m_2) \cdot m_1 m_2} \right)^2}}$$

D5

0.20 pts

Estimate the characteristic time τ of damping of the dumbbell oscillations.

Characteristic decay time of oscillations of order $\tau = \frac{1}{A}$:

$$\tau = \frac{(m_1 + m_2) \cdot m_1 m_2}{\gamma(m_1^2 + m_2^2)}.$$

Answer: Answer:

$$\tau = \frac{(m_1 + m_2) \cdot m_1 m_2}{\gamma(m_1^2 + m_2^2)}.$$

D6

0.20 pts

Determine the quality factor Q of such an oscillatory system.

We find the quality factor Q of this system using the well-known relation:

$$Q = \frac{\sqrt{B}}{A} = \sqrt{\frac{\Delta g}{m_1 m_2 l^2} \frac{(m_1 + m_2)^2 m_1 m_2}{\gamma \cdot (m_1^2 + m_2^2)}}$$

Answer: Answer:

$$Q = \sqrt{\frac{\Delta g}{m_1 m_2 l^2} \frac{(m_1 + m_2)^2 m_1 m_2}{\gamma \cdot (m_1^2 + m_2^2)}}$$

E1

0.10 pts

Calculate the moment of inertia I of the dumbbell rod about its axis.

As is known, the moment of inertia of a homogeneous solid cylinder relative to its axis of symmetry is equal to

$$I = mR^2/2. \quad (7)$$

E2

0.20 pts

Write down the law of change in the angular momentum of a dumbbell relative to its attachment point.

The derivative of the angular momentum of a rigid body with respect to time is ex-

pressed in terms of the total moment of external forces:

$$\dot{\vec{L}} = \vec{M}_{\text{out}}. \quad (8)$$

E3

0.50 pts

Calculate the angular velocity $\vec{\Omega}$ of the regular precession of the dumbbell.

$$\vec{L} = I\vec{\omega};$$

and in given case

$$\dot{\vec{L}} = I\dot{\vec{\omega}}.$$

Note, that torque тяжести equal $\vec{M}_{\text{out}} = \frac{m}{2} [\vec{l}, \vec{g}]$ and always direct perpendicularly $\vec{\omega}$ in horizontal плоскости. Thus it вызывает rotation vector $\vec{\omega}$ around the vertical.

Change in angular momentum over time interval Δt :

$$|\Delta \vec{L}| = L \sin(\theta) \Delta \varphi = M_{\text{out}} \Delta t.$$

Hence we get

$$\Omega = \frac{M_{\text{out}}}{L \sin(\theta)} = \frac{mgl}{2L}.$$

Answer: Answer:

$$\Omega = \frac{gl}{\omega R^2} = \frac{mgl}{2I\omega}.$$

F1

0.30 pts

Calculate the dependence of the dumbbell angular momentum modulus on time.

Note that the change in the modulus of the angular momentum of the dumbbell is affected only by the moment $\vec{M}_{\text{fric}}$, since all other moments of external forces are always directed perpendicular to $\vec{L}$.

$$\dot{L} = -\alpha\omega = -\frac{\alpha}{I}L \Rightarrow L = L_0 e^{-\frac{\alpha}{I}t}.$$

Answer: Answer:

$$L = L_0 e^{-\frac{\alpha}{I}t}.$$

F2

0.60 pts

Derive a differential equation describing the change in the angle of the dumbbell with the vertical over time.

Note that the moment caused by the force of viscous friction at the end m_2 is directed perpendicular to the dumbbell downwards and causes its axis to lower. That is, it causes an increase in the angle θ .

Let's calculate this moment:

$$M_2 = l \cdot \gamma \Omega l \sin(\theta).$$

Relate rotation moment momentum гантели with moment M_2 :

$$L\Delta\theta = M_2\Delta t = \gamma\Omega l^2 \sin(\theta)\Delta t.$$

Остаётся substitute here expression for Ω :

$$\dot{\theta} = \frac{\gamma l^2}{L} \frac{mgl}{2L} = \frac{\gamma mgl^3}{2L^2}.$$

Answer: Answer:

$$\frac{\dot{\theta}}{\sin(\theta)} = \frac{\gamma m g l^3}{2L_0^2} e^{\frac{2\alpha}{T}t}.$$

F3

0.20 pts

Let the starting angle be θ_0 . What is the rate of change of the angle at the initial moment?

Answer: Answer:

$$\dot{\theta} = \frac{\gamma m g l^2 \sin(\theta_0)}{2L_0^2}$$

F4

0.80 pts

With the initial condition from the previous paragraph, you obtain an explicit dependence $\theta(t)$.

The indefinite integral on the left is $\ln |\operatorname{tg}(\theta/2)| + C$, and on the right $\frac{\gamma m g l^2 I}{2\alpha L_0^2} \left(e^{\frac{2\alpha}{T}t} - 1 \right) + C$.
Integration limits: $\theta_0 \rightarrow \theta$, $0 \rightarrow t$.

Answer: Answer:

$$\theta = 2\operatorname{arctg} \left(\operatorname{tg} \left(\frac{\theta_0}{2} \right) e^{\frac{\gamma m g l^2 I}{4\alpha L_0^2} \left(e^{\frac{2\alpha}{T}t} - 1 \right)} \right)$$